



**Smart Social Media Trend Analysis & Content Management System**  
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**Subject: Data Structures and Algorithms (DSA–1)**

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# Problem Statement:-

Organizations need an efficient way to manage social media content, track engagement, identify trending topics, and analyze user interactions. Handling large amounts of content manually is difficult and time-consuming.

This project develops a Smart Social Media Trend Analysis & Content Management System using Data Structures and Algorithms (DSA) to store, search, manage, and analyze social media data efficiently while providing insights into content performance and audience reach.



# Objectives:-

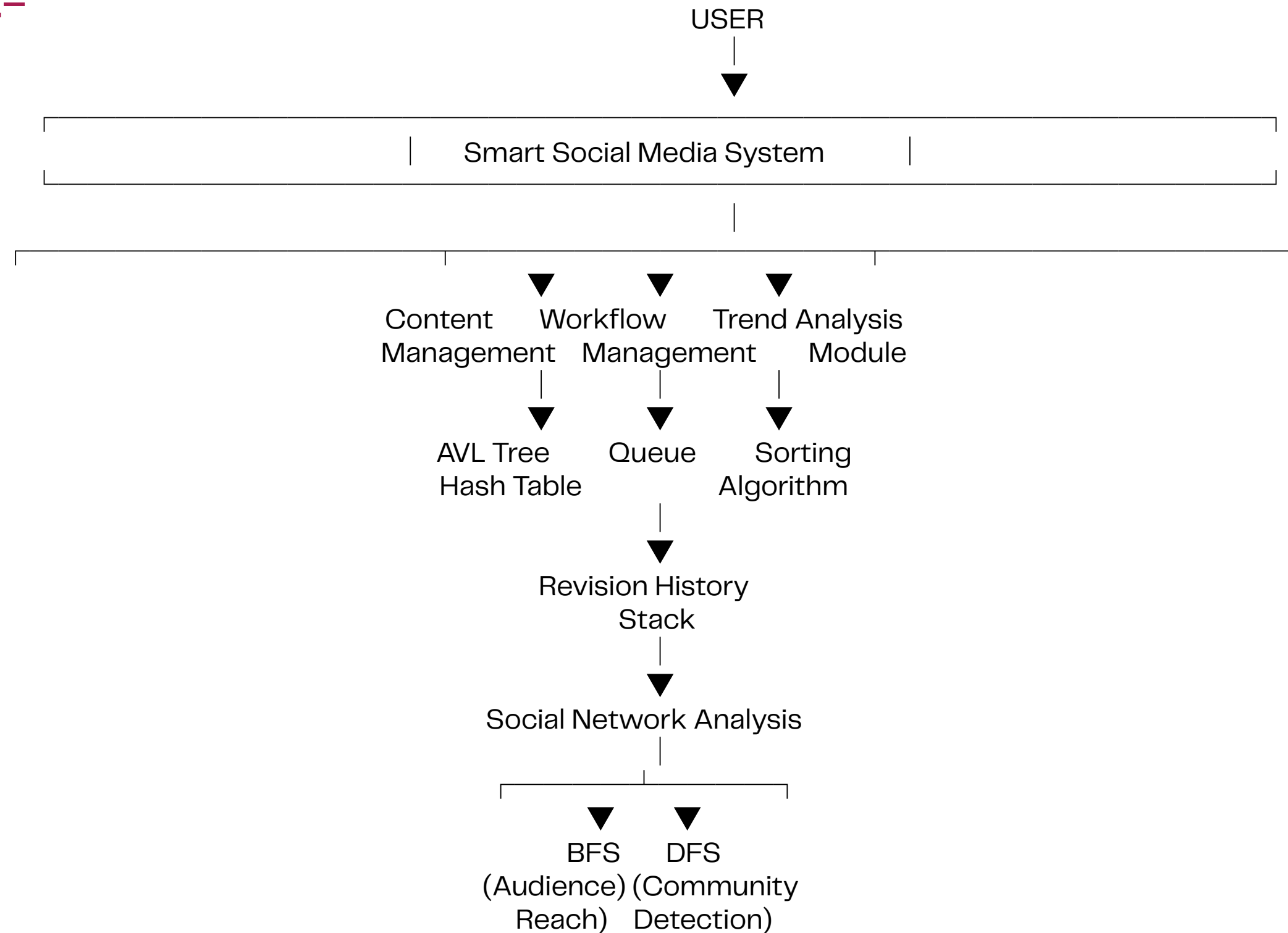
- Build a clear setup for handling social content.
- Sort trending posts using engagement numbers.
- Make storage and lookup fast with helpful data structures.
- Keep revision history with undo.
- Support an easy publishing flow.
- Study reach and user ties.
- Identify groups in networks.
- Show how AVL Trees, Hashing, Stack, Queue, BFS, and DFS work in practice.



# Design / Architecture:-

The Smart Social Media Trend Analysis & Content Management System is designed as an integrated platform that combines multiple Data Structures and Algorithms to efficiently manage social media content, analyze engagement trends, and study user interactions. The architecture is divided into six major modules that work together to provide a complete content management and analytics solution.

## Architecture Diagram:-



# Algorithm Description:-

| DSA Concept       | Purpose                             |
|-------------------|-------------------------------------|
| AVL Tree          | Store and organize content database |
| Hash Table        | Fast post retrieval                 |
| Stack             | Manage post revision history        |
| Queue             | Manage publishing workflows         |
| Sorting Algorithm | Rank trending posts                 |
| Graph             | Store user relationships            |
| BFS               | Audience reach analysis             |
| DFS               | Community detection                 |



# Complexity Analysis:-

## Time Complexity:-

| Operation     | Complexity    |
|---------------|---------------|
| AVL Insert    | $O(\log n)$   |
| AVL Search    | $O(\log n)$   |
| Hash Search   | $O(1)$        |
| Stack Push    | $O(1)$        |
| Stack Pop     | $O(1)$        |
| Queue Insert  | $O(1)$        |
| Queue Remove  | $O(1)$        |
| Sorting       | $O(n \log n)$ |
| BFS Traversal | $O(V + E)$    |
| DFS Traversal | $O(V + E)$    |

## Space Complexity:-

| Data Structure | Complexity |
|----------------|------------|
| AVL Tree       | $O(n)$     |
| Hash Table     | $O(n)$     |
| Stack          | $O(n)$     |
| Queue          | $O(n)$     |
| Graph          | $O(V + E)$ |



# Screenshots of Execution:-

```
ilghone29@SAHILs-Air-3 Smart Social Media Trend Analysis % ./SocialDashboard
[APH] Registered network member node: Alice Jenkins
[APH] Registered network member node: Bob Vance
[APH] Registered network member node: Charlie Green
[APH] Registered network member node: Diana Prince
[APH] Edge mapped. Connection built between User 1 and User 2
[APH] Edge mapped. Connection built between User 2 and User 3

=====
ENTERPRISE SOCIAL MEDIA TREND ANALYSIS & MGMT SYSTEM
=====

Add Post Instantly (AVL & Hash Entry)
Stage Content to Scheduling Queue (Queue Insertion)
Release Next Scheduled Workflow Item (Queue Removal)
Modify Existing Post Metadata (Stack Tracking)
Trigger Undo Command (Stack Pop Recovery)
Execute Rapid Lookup Index (O(1) Hash Map Search)
Render Performance & Campaign Dashboard (Sorting Engine)
Link Social Platform Users (Map Graph Matrix)
Compute Audience Spread Metrics (BFS Implementation)
  Scan Disjoint Market Clusters (DFS Strategy)
Diagnostic: Output In-Order Structure Sequence
Shutdown Core Engine Processing Framework

=====
Enter Executive Action Selection Index: █
```

```
-----
Enter Executive Action Selection Index: 7

=== PERFORMANCE ANALYTICS DASHBOARD (Sorted by Engagement Score) ===
Rank Index -> Score: 260 | [Category: Test] Title: Tech Post (ID: 999)
[STRATEGY]: Prioritize High-Engagement Campaign Resources & Targeting Options.
Rank Index -> Score: 120 | [Category: Business] Title: Queue Test (ID: 200)
[STRATEGY]: Prioritize High-Engagement Campaign Resources & Targeting Options.
```

```
-----
Enter Executive Action Selection Index: 8
Source ID: 4
Destination Target ID: 5
[GRAPH] Edge mapped. Connection built between User 4 and User 5
-----
```

```
-----
Enter Executive Action Selection Index: 1
Set unique numeric Post ID: 999
Set Content Title: Tech Post
Set Analytical Category: Test
Input Baseline Metrics [Likes Comments Shares]: 100 50 20
[DATABASE] Post record added to AVL database & Hash storage.
```

```
-----
Enter Executive Action Selection Index: 9
Enter Base Search Root Node: 1

[BFS AUDIENCE INSIGHTS SPREAD TRAVERSAL]:
-> Alice Jenkins -> Bob Vance -> Charlie Green
[ANALYTICS SUMMARY] Total Content Audience Reach Matrix Count: 3 users.
```

```
-----
Enter Executive Action Selection Index: 2
Set Target Task Post ID: 200
Set Queue Task Title: Queue Test
Set Category Tag: Business
Project Target Benchmarks [Likes Comments Shares]: 50 20 10
[QUEUE] Post staged into the content management buffer pipeline.
```

```
-----
Enter Executive Action Selection Index: 10

[DFS NETWORK ISOLATED COMMUNITY CLUSTERING DETECTION]:
Cluster Profile Group 1: [ (ID: 5) Diana Prince (ID: 4) ]
Cluster Profile Group 2: [ Charlie Green (ID: 3) Bob Vance (ID: 2) Alice Jenkins (ID: 1) ]
```

```
-----
Enter Executive Action Selection Index: 3
[WORKFLOW] Dequeuing and parsing post validation parameters...
[DATABASE] Post record added to AVL database & Hash storage.
```

```
-----
Enter Executive Action Selection Index: 11

=== Raw Database Contents (BST In-Order Display) ===
[ID: 200] Title: Queue Test | Engagement: 120
[ID: 999] Title: Tech Post | Engagement: 260
```

```
-----
Enter Executive Action Selection Index: 4
Enter Target Post ID to edit: 999
Enter Revised Content Title: Updated Test Post
Enter Revised Category Tag: Technology
[STACK] Revision snapshot saved. Post updated successfully.
```

```
-----
Enter Executive Action Selection Index: 5
[UNDO] Success. Content rolled back to previous version state.
```

```
-----
Enter Executive Action Selection Index: 6
Target Index Query ID: 999

[HASH SEARCH RESULT - O(1)]
ID: 999
Title: Tech Post
Category: Test
Engagement Score: 260
```

```
-----
Enter Executive Action Selection Index: 0
De-allocating dynamic memories. Execution pipeline stopped safely.
%
sahilghone29@SAHILs-Air-3 Smart Social Media Trend Analysis % █
```



## Results and Observations:-

- Successfully implemented all required DSA concepts.
- AVL Tree provided efficient and balanced post storage.
- Hashing enabled fast post retrieval and searching.
- Stack maintained revision history with undo functionality.
- Queue managed content scheduling and publishing workflows.
- Sorting ranked posts based on engagement scores.
- BFS analyzed audience reach across the social network.
- DFS identified connected user communities.
- The system effectively tracked content performance and trending posts.



## Conclusion:-

The Smart Social Media Trend Analysis & Content Management System was successfully implemented using C++ and Data Structures & Algorithms.

The project integrates AVL Trees, Hash Tables, Stack, Queue, Sorting Algorithms, Graphs, BFS, and DFS into a single application that performs content management, trend analysis, audience reach tracking, and community detection.

The system demonstrates how DSA concepts can be applied to solve real-world social media management problems efficiently. The project fulfills all requirements of PS 127 and provides an effective platform for analyzing content performance and user engagement.



## GitHub Repository Link:-

Project Repository:

<https://github.com/sahilghone29-cell/-Smart-Social-Media-Trend-Analysis-Content-Management-System.git>

Repository Owner: Sahil Ganesh Ghone

Purpose: Contains the complete source code, README documentation, screenshots, and project files for the Smart Social Media Trend Analysis & Content Management System developed using C++ and Data Structures & Algorithms (DSA).

